Angular Advanced -
Smart components & View components

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“In Angular, all components are equal. There is no sense of different ‘types’ of components”

Smart components / View Components

- Design pattern
- Why ? Separation of concerns
  - View component is responsible for presentation (and *can* be used in a completely different environment with different component logic)
  - Smart component is responsible for logic → passing the [calculated] data to the view component.
- Smart components
  - AKA: *Container* components, *controller* components, *statefull* components
- View components
  - AKA: *pure* components, *dumb* components, stateless

Characteristics

- Smart components
  - Typically contain big(ger) chunks of logic
  - Typically have a small UI part (reference just the view component)
  - Pass the data to the view component
- View components
  - Typically contain no logic whatshowever
  - Get their data via `@Input()` decorators
  - Submit events via `@Output()` decorators
  - Have large chunks of UI

Example: ../300-smart-view-component

Home (default component) | Smart/View component

Default component- My favorite cities:

1 - Groningen	☑ Visited
2 - Hengelo	☐ Visited
3 - Den Haag	☐ Visited
4 - Enschede	☑ Visited
5 - Heerlen	☐ Visited

I Visited:

Groningen

Enschede

Current city: Groningen

Home – Default component

- Contains all logic and UI, combined in one component.
  - `/home/home.component.html`
  - `/home/home.component.ts`
- This typically works well for smaller applications

```
<div class="row">
   <div class="col-md-6">
      <h2>Default component- My f
      <ul class="list-group">
         <li *ngFor="let city of
            class="list-group-ite
            [class.visited]="city
            (click)="getCity(city
            {{ city.id}} - {{ cit
            <span class="pull-rig
               <label>
      </ul>
..
</div>
```

```
@Component({
   selector    : 'app-home',
   templateUrl: './home.component.html'
})
export class HomeComponent implements OnInit {
   …
   ngOnInit() {
      this.cityService.getCities()
         .subscribe(cities => this.cities = cities);
   }

   getCity(city: City) {
      this.currentCity = city;
      this.cityPhoto   = `assets/img/${this.currentCity.name}.jpg`;
   }
   …
}
```

Splitting it up in view components

- `../smart-view/smart.component.html | .ts.`
- Passing `[cities]` in
- Getting `(events)` out.
- `[cities]` can also be an Observable

```
<div class="row">
  <div class="col-md-6">
    <h2>Smart/view component- My favorite cities:</h2>
    <!-- The <city-list>-component is now
         the view component for list of cities -->
    <city-list [cities]="cities"
               (selectCity)="getCity($event)"
               (toggleVisited)="toggleCity($event)">
    </city-list>
   …
  </div>
</div>
```

<city-list> View component

- Has no logic, just `@Input()`'s and `@Output()`'s

```
import {Component, EventEmitter, Input, Output} from '@angular/core';
import {City} from '../../shared/model/city.model';

@Component({
   selector   : 'city-list',
   templateUrl: './city-list.component.html'
})
export class CityListComponent {

   @Input() cities: City[];
   @Output() selectCity: EventEmitter<City>    = new EventEmitter<City>();
   @Output() toggleVisited: EventEmitter<City> = new EventEmitter<City>();

}
```

HTML of the view component

- …/`smart-view/city-list/city-list-component.html`
- Has a nested view component to toggle `visited` state
- Choice: events are directly emitted from the HTML
  - Can also be done via small functions

```
<ul class="list-group">
   <li *ngFor="let city of cities"
      class="list-group-item"
      [class.visited]="city.visited"
      (click)="selectCity.emit(city)">
      {{ city.id}} - {{ city.name }}
      <city-visited [visited]="city.visited"
                  (toggle)="city.visited = $event; toggleVisited.emit(city)">
      </city-visited>
   </li>
</ul>
```

`<city-visited>` View component

Again – just `@Input()`'s and `@Output()`'s.

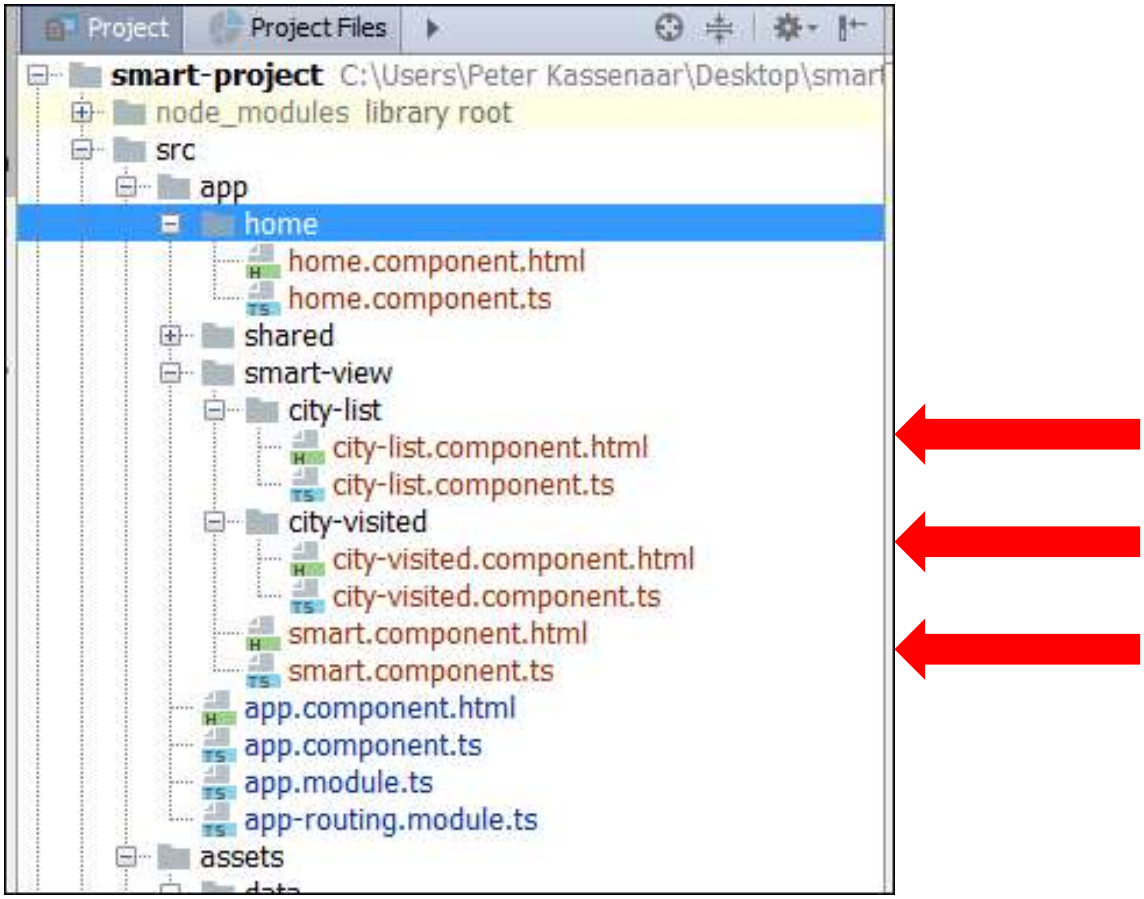
```
export class CityVisitedComponent {
  @Input() visited:boolean;
  @Output() toggle: EventEmitter<boolean>= new EventEmitter<boolean>();
}
```

HTML – again: attribute binding and event binding

```
<span class="pull-right">
   <label>
      <input type="checkbox"
            class="checkbox-inline"
            [checked]="visited"
            (change)="visited = !visited; toggle.emit(visited)"> Visited
   </label>
</span>
```

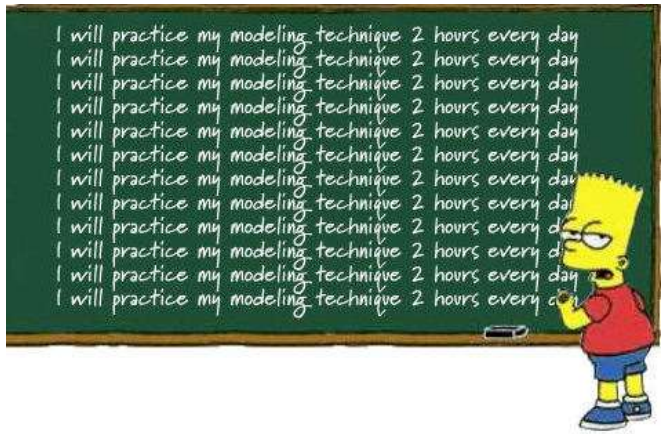
Application structure

- When following this pattern: typically more, smaller components
- The directory tree gets crowded!

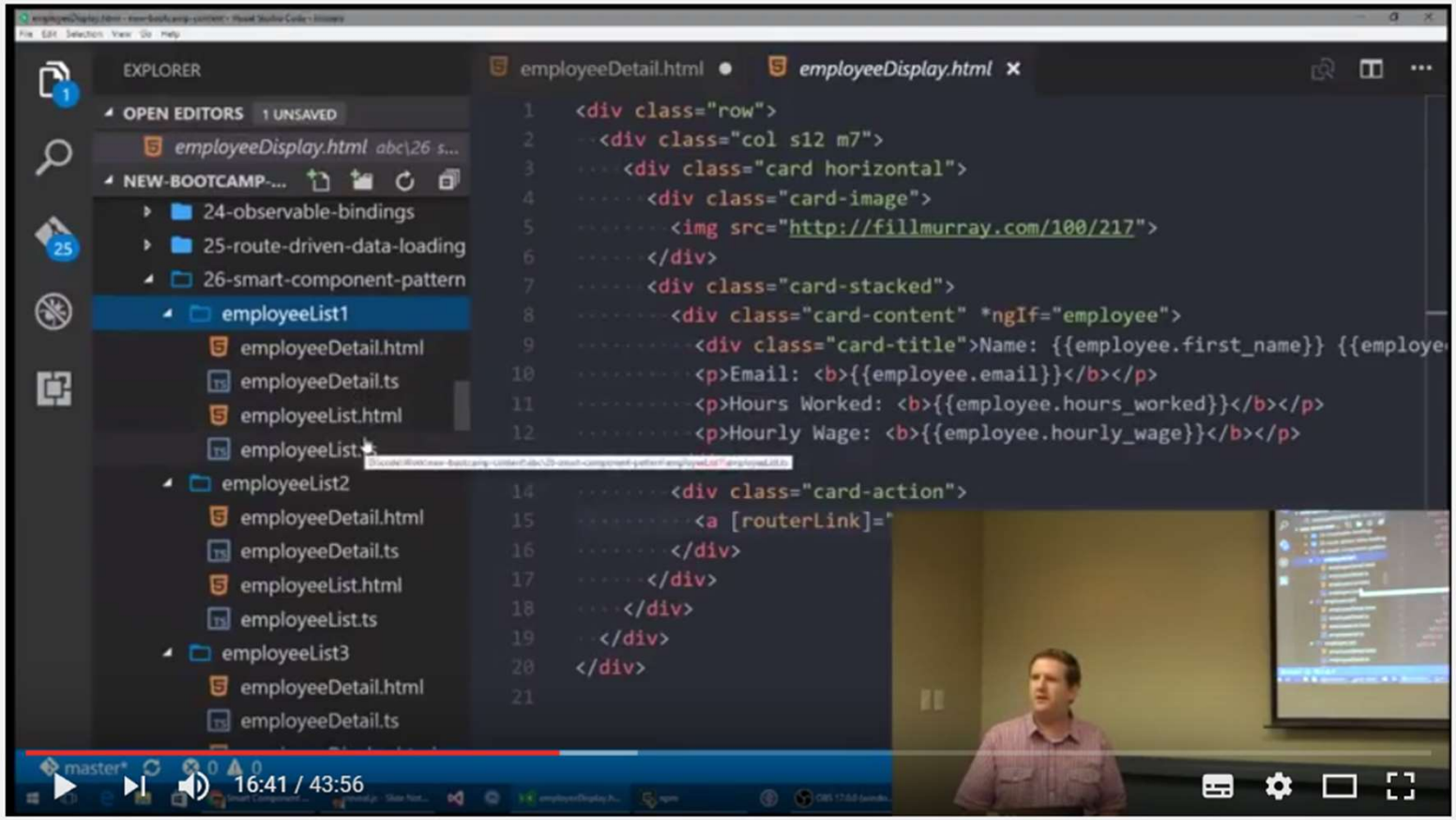


Workshop

- Study the example `../300-smart-view-components`
- Create two additional view components:
  - One for displaying the current city
  - One for displaying the list of visited cities
- Optional: use Observables instead of plain arrays.
  - `cities$ : Observable<City[]>`
  - `[cities]="cities$ | async"`



More info on Smart/View components



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Stateful and stateless components, the missing manual

Oct 12, 2016 10 mins read Edit post

The goals of this article are to define what stateful and stateless components are, otherwise known as smart and dumb - or container and presentational components. For the purposes of the article, we'll be using Angular 2 Components to explain the stateful/stateless concepts. Bear in mind these concepts are not at all limited to Angular, and live in other libs/frameworks such as React.

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https://toddmotto.com/stateful-stateless-components

http://blog.angular-university.io/angular-2-smart-components-vs-presentation-components-whats-the-difference-when-to-use-each-and-why/

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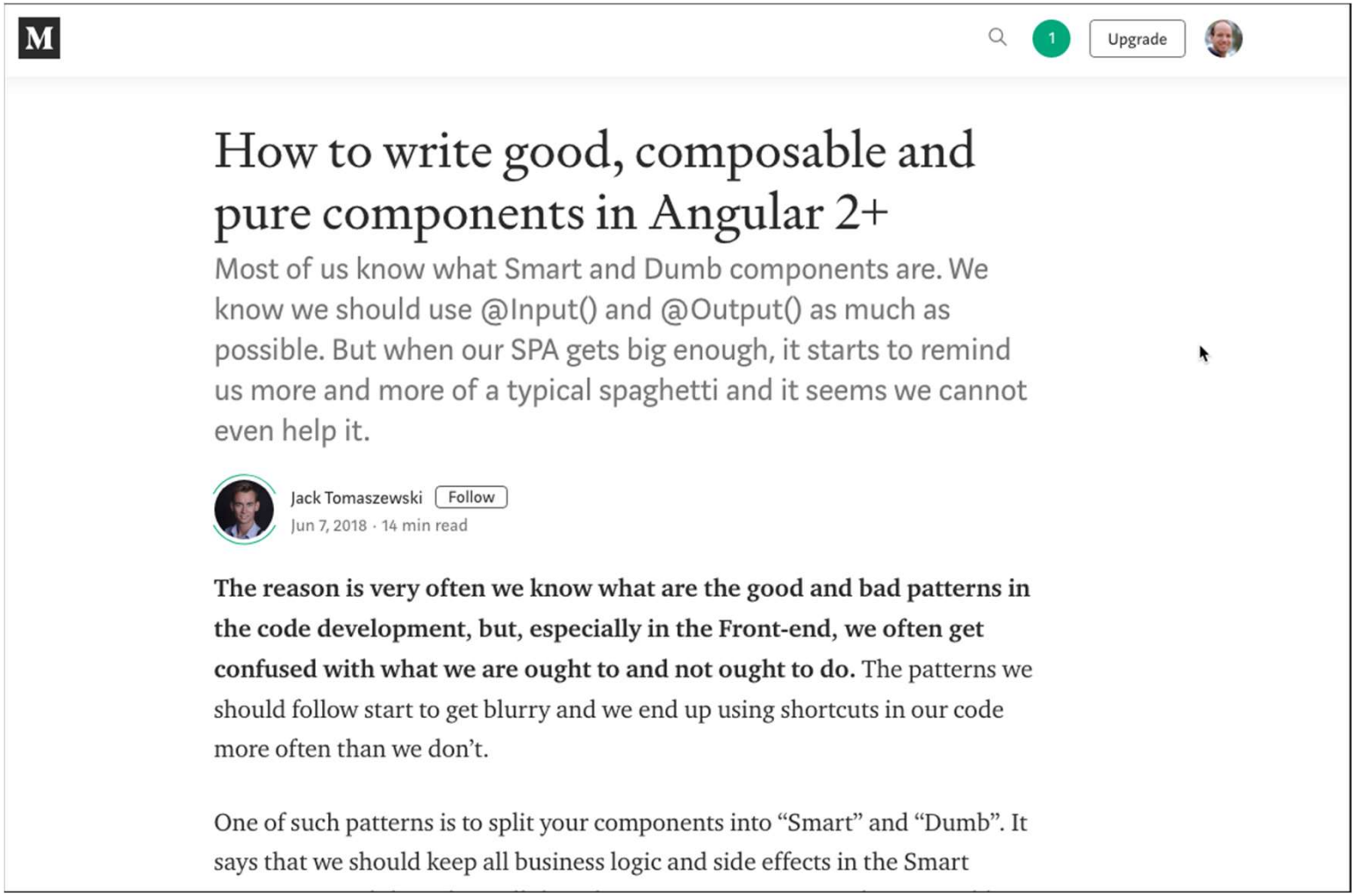
https://medium.com/@dan_abramov/smart-and-dumb-components-7ca2f9a7c7d0



How to write good, composable and pure components in Angular 2+

Most of us know what Smart and Dumb components are. We know we should use @Input() and @Output() as much as possible. But when our SPA gets big enough, it starts to remind us more and more of a typical spaghetti and it seems we cannot even help it.

Jack Tomaszewski Follow
Jun 7, 2018 · 14 min read

The reason is very often we know what are the good and bad patterns in the code development, but, especially in the Front-end, we often get confused with what we are ought to and not ought to do. The patterns we should follow start to get blurry and we end up using shortcuts in our code more often than we don't.

One of such patterns is to split your components into "Smart" and "Dumb". It says that we should keep all business logic and side effects in the Smart

https://medium.com/@jtomaszewski/how-to-write-good-composable-and-pure-components-in-angular-2-1756945c0f5b